$$\frac{\Gamma \vdash e_1 : \tau \to \mathsf{bool} \rightarrowtail e_3 \qquad \Gamma \vdash e_2 : \tau \rightarrowtail e_4}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) : \tau \rightarrowtail \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_3, e_4 \rangle\rangle}\ \text{Compile-Flat}$$

$$\frac{\Gamma \vdash e_1 : \tau_2 \rightarrowtail e_2}{\Gamma \vdash \lambda f : \tau_1 \to \tau_2 . \lambda x : \tau_1 . \mathsf{mon}_j^{k,l}\Big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail e_3}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail e_3\ e_2}\ \text{Compile-Func}$$

$$\frac{\begin{array}{c} f, x \notin \mathrm{dom}(\Gamma) \cup \{y\} \cup \mathrm{fv}(\kappa_1, \kappa_2) \qquad \Gamma \vdash e_1 : \tau_1 \to \tau_2 \rightarrowtail e_2 \\ e_w = \lambda f : \tau_1 \to \tau_2 . \lambda x : \tau_1 . \mathsf{mon}_j^{k,l}\Big(\kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big], f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\ \Gamma \vdash e_w : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail e_3 \end{array}}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y . \kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail e_3\ e_2}\ \text{Compile-Dep}$$

$$\frac{\Gamma \vdash e_1 : \tau \to \mathsf{bool} \rightarrowtail^l e_3 \qquad \Gamma \vdash e_2 : \tau \rightarrowtail^l e_4}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) : \tau \rightarrowtail^l \mathtt{handler}_s(\mathtt{h}_{\mathscr{C}})\ \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_3, e_4 \rangle\rangle}\ \text{Local-Compile-Flat}$$

Figure 1: Compiler Rules